
Who Hallucinates Tools, How Often, and What Fixes It: A Per-Call Study Across Anthropic 4.x and Qwen3

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Abstract

Tool-name hallucination is a known failure mode. Who actually does it, how often, and what fixes it? We define a per-call Tool Fabrication Rate (TFR) and run it on BFCL multi-turn against two populations: five Anthropic 4.x versions across an eleven-month release window, and the Qwen3-Instruct family at six sizes from 0.6B to 32B (4-bit MLX). The Anthropic side is quiet. Across 2,599 opaque-baseline calls we logged zero TFR events, Clopper-Pearson 95% upper bound $\leq 0.115\%$. Commercial models across the OpenAI gpt-4.1 and gpt-4o tiers also log 0% TFR (0/2,070 pooled, $\leq 0.14\%$). Qwen3 isn't: its rate is non-monotone in scale, climbing from 0% at 0.6B to **1.87%** at 14B and back to 0% at 32B. We propose Registry-Visible Reprompting (RVR), a training-free middleware that, on a registry miss, returns one re-prompt wrapping a structured `tool_not_found` envelope. RVR removes all fourteen pooled Qwen3 fabrications (Fisher's exact one-sided, $p = 7.1 \times 10^{-5}$ on 14/973 vs. 0/945). The surprise is where the fix lives. An ablation that keeps the envelope but drops the registry listing ($C_{0.7}$; 0/448 on Qwen3-8B, Clopper-Pearson 95% upper bound $\leq 0.67\%$) already hits zero, and a decoy that fills the envelope with wrong tool names ($C_{0.8}$) does not re-introduce fabrications¹. The recovery is format, not content: a single structured rejection re-grounds the model regardless of what the registry says, echoing the format-over-content effect Min et al. found for in-context demonstrations (Min et al., 2022). Read this way, the mid-scale bump stops being noise and becomes a pre-

dicted signature: a U-shaped capability window (Wei et al., 2022; McKenzie et al., 2023) in the generative tool-name prior, controlled for quantization by the 0-and-32B endpoints that bracket it at the same 4-bit precision. On Anthropic's zero-event regime RVR also logs zero, but the strict-pass subset slips at $N = 60$; we recommend tier-conditional deployment. Our read: reactive agent error-recovery is less about telling the model the right answer than about re-grounding its output format, and one structured reply does it on the models where the failure shows up.

1. Introduction

An agent calls `search_flights`. The registry contains `find_flights`. The call fails. Practitioners hit this mismatch in production with Claude, Gemini, and Grok (Czapla, 2026), and what stings is that the model knew the task, picked something plausible, and just got the name wrong. The reflex fix is to hand the model the registry: show it the menu and it will order off it. We find the menu is mostly decoration. What clears the failure is the shape of the rejection, not its contents: a structured `tool_not_found` envelope re-grounds the model whether or not it lists a single real tool. We call this format-not-content recovery, and it is the spine of this paper.

The framing has a pedigree. Min et al. showed that in-context demonstrations help even when their labels are wrong: the format carries the signal, the content rides along (Min et al., 2022). We report the reactive-recovery analogue: a registry-miss re-prompt fixes tool-name fabrication even when the listed registry is empty ($C_{0.7}$) or actively wrong ($C_{0.8}$). Existing benchmarks measure related quantities at the wrong grain, per-task or per-query and folded into composite scores (Li et al., 2023; Huang et al., 2024; Xu et al., 2024; Section 2). Read together, the failure looks architecture-agnostic and uniform. We don't think it is, and we don't think the fix is what it looks like either.

¹The decoy arm logs zero TFR across all four Qwen3-8B probe arms (0/410, $\leq 0.73\%$: `near_name` 0/97, `synonym` 0/106, `matched_random` 0/107, `unrelated` 0/100), matching $C_{0.7}$ (0/448) and the real list C_1 (0/258); registry content is decorative for recovery.

We define the per-call Tool Fabrication Rate (TFR) and run it over two populations on BFCL multi-turn: the Anthropic 4.x family at five versions spanning May 2025 to April 2026, and Qwen3-Instruct at six sizes (0.6B–32B) in 4-bit MLX on consumer hardware. Both see three BFCL splits plus a four-distractor probe with the real target removed and a synthetic look-alike in its slot. BFCL is strictly typed, so this isn’t production, but it is the closest we could get to Czapla’s setting on a benchmark. The Qwen3 scaling curve (Table 3, plotted in Appendix Figure 2) is the headline.

Why a per-call existence rate matters: a fabricated name is not just a failed call, it is an attack surface. Spracklen et al. (2025) audited package hallucinations in code-generating LLMs and found open-source models invent non-existent dependencies at $\sim 21.7\%$ against $\sim 5.2\%$ for commercial ones, an open-vs-closed gap that mirrors the Qwen-vs-Anthropic split we report and that an adversary can turn into “slopsquatting” by registering the hallucinated name and waiting for the model to reach for it. Tool registries are the same shape of target. We make no exploitation claim, and measure existence hallucination at the per-call grain because that is the unit at which such a risk would accrue.

The two families don’t look alike. Across 2,599 Anthropic calls under C_0 , zero TFR events (Clopper–Pearson 95% upper bound $\leq 0.115\%$). The five commercial OpenAI gpt-4.1 and gpt-4o tiers also log 0% TFR (0/2,070 pooled, $\leq 0.14\%$). The split designed to coax fabrication, `multi_turn_miss_func`, didn’t bite: the agent chained five smaller in-registry calls instead of inventing a one-shot. With every Anthropic cell at $0/N$, no temporal slope is fittable; the five versions function as a stability check, not a trend.

Qwen3 looks nothing like that. Pooled per-call rates climb from 0% at 0.6B, peak at **1.87%** at 14B, and drop back to 0% at 32B. The bump is not noise. A generative tool-name prior that strengthens with scale before a larger model’s registry-grounding control catches up predicts exactly this shape: a U-shaped capability window (Wei et al., 2022; McKenzie et al., 2023). The 0% endpoints at 0.6B and 32B bracket the 1.87% peak at the same 4-bit MLX precision, so quantization is held fixed across the arc; the curve is a capability effect, not a compression artifact. Six points still isn’t a curve, and we say so, but it is the shape the format-not-content account predicts.

Registry-Visible Reprompting (RVR) is what we built to fix this. The middleware is training-free: intercept the proposed call, check the name, and on a miss re-prompt with a structured `tool_not_found` envelope

listing what is available. Across the four Qwen3 tiers with non-zero TFR, RVR clears all fourteen pooled fabrications: Fisher’s exact one-sided on 14/973 vs. 0/945 gives $p = 7.1 \times 10^{-5}$. A content-matched ablation that drops the registry list ($C_{0.7}$) also hits zero, and a decoy that fills the envelope with wrong names ($C_{0.8}$) still logs zero. The envelope is doing the work, not the listed names. On the Anthropic zero-event regime RVR holds at zero, but strict-pass slips (Sonnet 60 $\rightarrow$ 53/60; Haiku 60 $\rightarrow$ 56/60); a Newcombe-hybrid TOST at $N = 60$ is underpowered for a 5-pp non-inferiority margin, so we can’t rule the cost in or out. We argue for tier-conditional deployment.

The Anthropic C_0 bound ($\leq 0.115\%$ at $N = 2,599$) sits an order of magnitude below the Qwen3 point estimates in the 0.95–1.87% band. The bound-vs-estimate asymmetry is the finding. The intra-family non-monotonic shape on Qwen3 is itself descriptive evidence aggregate metrics couldn’t have surfaced. What we give a deployer is concrete: (i) a per-call diagnostic that exposes existence failures production guardrails silently absorb, (ii) the format-not-content finding (recovery rides on the rejection envelope, not the registry it echoes), and (iii) the deployable takeaway that falls out of it: ship the structured envelope, and you need not leak your registry to get the fix. We release four artifacts: the per-call TFR diagnostic, an eleven-model audit over eleven months, RVR with per-tier validation, and an open-source harness with 144 unit tests. Total API spend \$73; the Qwen3 sweep ran at zero marginal cost on the M5.

2. Related Work

Tool-using agents and benchmarks. Tool-using LLM agents are evaluated on a small, fairly stable set of suites. The Berkeley Function-Calling Leaderboard (Patil et al., 2024; 2025) started as single-turn API selection and has grown into a stateful multi-turn suite. τ -bench (Yao et al., 2024) pushes in a different direction, scoring dialog-driven retail and airline agents that must reconcile a user goal against a tool catalog. Both report end-to-end task success, which fuses tool selection with reasoning, parsing, and policy errors into one number. Trace-level harnesses (Trivedi et al., 2024; Yao et al., 2023) expose more, yet still report pass-rate as the headline. This is the wrong granularity for the failure we care about; TFR isolates a portable slice of it across vendors and benchmarks.

Empirical analysis of agent failures. Practitioners have informally reported that frontier models call tools that were never declared. Czapla (2026) reproduces

the behavior across Claude, Gemini, and Grok, noting that “Claude 4.5 hallucinated access to a tool I hadn’t given it yet ...I’ve reproduced similar behavior with Gemini and Grok.” On the academic side, [Zhu et al. \(2026\)](#) mine 998 bug reports from CrewAI and LangChain and find “API misuse,” “API incompatibility,” and “Documentation Desync” dominating the root-cause distribution, with the Self-Action lifecycle stage hit hardest. Neither line puts a number on the failure under controlled conditions. We do, formalizing the unauthorized-tool subclass as TFR and reporting it across five models.

Prior tool-hallucination metrics. Tool-call name errors have been measured under several aggregations, and the choice of denominator matters. [Li et al. \(2023\)](#) introduce “API Hallucination” in API-Bank as a ground-truth-vs-prediction name-mismatch error category, with per-task error rates up to 61.4% on then-current models. The closest prior measurement to ours is [Huang et al. \(2024\)](#), whose MetaTool subtask 3 deliberately removes the correct tool from the registry and tracks the resulting fabrication through a per-query Correct Selection Rate. That isolates the registry-membership phenomenon cleanly, but stays at the per-query level on single-turn instructions. [Xu et al. \(2024\)](#) name the phenomenon directly (“tool type hallucination,” where the model fabricates a tool absent from the available set), but their fix is a training-time reliability alignment that expands the action space (defer / clarify / switch), and they report it inside a composite reliability score rather than as a standalone per-call rate. We share their target failure but differ on both axes: a per-call rate, and a training-free inference-time intervention. The “does the called tool exist” question is not itself new: ToolBeHonest ([Zhang et al., 2024](#)) catalogues a non-existent tool error category qualitatively, and BFCL v4’s hallucination metric ([Patil et al., 2025](#)) measures a different failure, invoking a tool when none is appropriate. What is new is the per-call operationalization across capability tiers and vendors, and the content-controlled ablation (Section 6) it makes possible: TFR extends MetaTool’s premise from per-query to per-call on multi-turn agentic traces, and from controlled removal to the natural distribution of registry-membership violations. Per-call is what lets the intervention in Section 3.2 be evaluated against a content-controlled $C_{0.7}$ baseline, a contrast aggregate metrics cannot expose. Appendix Table 5 positions TFR against these prior measurements by what each one counts; the denominators and elicitation setups differ, so the numbers are not directly comparable, but the granularity gap is the point.

Constrained decoding and self-correction. A separate line constrains agents at decode time. Grammar-constrained and JSON-mode decoding ([Willard & Louf, 2023](#); [OpenAI, 2024a](#)) prevent malformed calls. Provider-side function-calling APIs reject unknown names but surface the error as an opaque exception ([OpenAI, 2024b](#); [Anthropic, 2024](#)). Self-correction methods such as Reflexion ([Shinn et al., 2023](#)) and self-debug ([Chen et al., 2023](#)) push the diagnosis back onto the agent’s own trace. The closest neighbor in method shape is CRITIC ([Gou et al., 2024](#)), an external-feedback verify-then-correct loop that needs no training and treats the model as a black box. RVR is a registry-membership-specialized instance of that pattern. What distinguishes it is the content-controlled $C_{0.5}/C_{0.7}/C_1$ ablation, which separates whether the gain comes from retrying, from a structured-error code, or from the registry list itself. Production frameworks have started to adopt structured tool-not-found feedback; the LangChain community ([limerickgds, 2025](#)) explicitly proposes returning a structured ToolMessage on a failed call, but stops short of echoing the available registry. That gap is exactly what our $C_{0.7}/C_1$ contrast measures.

Closely related interventions. Three concurrent works deserve direct comparison. [Kumar & Cohen \(2026\)](#) introduce Localized In-Context Learning (L-ICL) for symbolic planners, locating the first constraint violation and injecting a minimal corrective example, lifting valid-plan rate from 59% to 89% on 8×8 gridworld. RVR is similarly minimal but operates on tool-using agents and pulls its signal from the runtime registry rather than curated examples. [Zhang et al. \(2026\)](#) train Qwen3-8B with Fission-GRPO, an RL recipe that resamples failed trajectories with diagnostic feedback, gaining +5.7pp error recovery on BFCL-v4 multi-turn. That is a training-time fix on the same benchmark family we target at inference time, and the two are orthogonal. [Engländer et al. \(2026\)](#) describe a related failure: agents discover useful signals but fail to act on them (“agents use the environment to fetch expected information, but not to revise their strategy”). RVR addresses the dual problem of acting on unobserved tools, evaluated as a paired content-controlled contrast rather than observationally.

Adjacent work. [Guo et al. \(2024\)](#) targets tool-server stability rather than name existence, and production tool-skill packaging ([Anthropic, 2025](#)) is orthogonal.

Concurrent work on tool hallucination (2026). Three 2026 papers bracket our setting. [Yin et al. \(2026\)](#) in-

Introduce SimpleToolHalluBench, whose “no tool available” and “only distractor tools” regimes mirror our target-removed probe, and report that strengthening reasoning (RL/SFT/prompting) monotonically amplifies tool hallucination. Qi (2026) find a non-monotonic effect on a different axis, chain-of-thought budget at fixed scale on Qwen2.5-1.5B function calling. Our axis is orthogonal to both: model scale at fixed (disabled) thinking, where the non-monotonicity appears across the Qwen3 size ladder rather than along a reasoning knob. Healy et al. (2026) detect tool-selection hallucination from a model’s internal representations in a single forward pass; we work purely behaviorally and, rather than detect, prevent with a message-level re-prompt that needs no logit or activation access.

3. Method

3.1. Tool Fabrication Rate (TFR)

Let a tool registry $\mathcal{R}$ be a finite set of tool names with JSON schemas, exposed to the agent at episode start. The agent’s message stream is parsed into a sequence of tool calls c , each carrying a name $c.name$, and we define

$$\text{TFR}(m, b, x) = \frac{|\{c : c.name \notin \mathcal{R}\}|}{|\{c : c \text{ is a parsed tool call}\}|},$$

where m, b, x index model, benchmark, and condition. The denominator is per call, not per task. Per-task aggregation conflates the rate with trace length: on long tool-using episodes, a single bad agent emitting twenty bad calls swings the headline number by an order of magnitude relative to one. Per-call observations inside a task are not assumed i.i.d.; task-level cluster-bootstrap confidence intervals are reported instead. System failures (timeout, refusal, parse-fail) are excluded from numerator and denominator alike.

3.2. Registry-Visible Reprompting (RVR)

RVR is a single-turn intervention placed between the agent and the tool executor (Appendix Figure 3). On a membership-violating call, it returns a re-prompt that names the offending call and lists the available tools:

```
def rvr(agent_msg, registry):
    proposed = parse_tool_call(agent_msg)
    if proposed.name not in registry:
        return Action.RE_PROMPT(
            f"Tool '{proposed.name}' not in registry. "
            f"Available: {sorted(registry.keys())}.")
    return Action.EXECUTE(proposed)
```

Echoing $\mathcal{R}$ back into the conversation places valid names in the model’s recent context, which is what

the next decoding step attends to. The intervention uses one re-prompt, a production-realistic budget; it is training-free and assumes no gradient access; it composes with RL-based methods such as Fission-GRPO (Zhang et al., 2026); and it is the message-level analog of constrained decoding for closed-API regimes without logit access.

3.3. Conditions and the content-vs-format question

Five conditions are run on a hallucinated call. C_0 is the opaque baseline, returning the framework’s default error string. $C_{0.5}$ is a naive retry with the literal text “previous call failed, try again”. $C_{0.7}$ is a structured error, JSON `{"error": "tool_not_found"}`, with no registry attached. C_1 is RVR proper (Section 3.2): the structured envelope plus the sorted real registry. $C_{0.8}$ holds the C_1 envelope prose verbatim but swaps the listed tools for a count-matched decoy set guaranteed absent from the registry. The ladder is built so that adjacent steps isolate one factor each: $C_{0.7} : C_{0.5}$ isolates structure, $C_1 : C_{0.7}$ isolates the presence of a list, and $C_1 : C_{0.8}$ isolates whether the listed names must be correct. At scale we report the conservative $C_1 : C_0$.

We do not try to localize the effect to a specific cognitive story. An earlier draft pre-registered three mechanisms (induction-head copying, registry-as-cue, message-level rejection); the data cannot separate copying from cueing, so we drop that apparatus rather than over-read it. What the design can decide is a single binary: does recovery require the registry content, or only the rejection format? If a wrong list ($C_{0.8}$) recovers as well as the right list (C_1), the content is decorative; if it fails where C_1 succeeds, the model is reading the offered names. The distractor probe (Section 3.4) is retained to characterize where on the scale curve fabrications concentrate, not as a mechanism discriminator.

3.4. Distractor-Type Probe

One distractor is injected into $\mathcal{R}$ on tasks where C_0 already passes, with its type varied to characterize which kind of look-alike concentrates fabrications (Appendix Figure 4). `near_name` draws a string at Levenshtein distance 1 or 2 from a real target. `synonym` is high embedding-cosine but Levenshtein-far: semantic but not orthographic overlap. `matched_random` is lexically random, matched to the real target on description length and arity, controlling for the “registry-size +1” confound. `unrelated` is lexically and semantically disjoint, the floor case. To capture the Czapla-style runtime collision, the real target is also removed and the distractor placed in its slot: the agent expects a

tool that exists, but the registry offers only a similarly-named substitute.

4. Experimental Setup

Models. We evaluate two populations. The Anthropic 4.x family (Opus 4.7, Sonnet 4/4.5/4.6, Haiku 4.5) is queried through the provider’s tool-use API at temperature=0, with rolling aliases resolved at run-start to dated snapshots and pinned in the manifest. The Qwen3-Instruct family (0.6B, 1.7B, 4B, 8B, 14B, 32B) runs locally as mlx-community 4-bit MLX builds on an Apple M5 with 32 GB unified memory, macOS 15. The manifest lists SDK and dataset pins. A supplementary cross-family release covers six additional open models: Llama-3.1-8B, Mistral-7B-v0.3, Gemma-2-9B, Qwen2.5-7B, Phi-3.5-mini, and DeepSeek-R1-Distill-Qwen-7B. Both populations decode greedy.

Benchmark. All experiments run on BFCL v4 multi-turn, using three splits: `multi_turn_base` (aligned registry), `multi_turn_miss_func` (one required tool removed), and `multi_turn_long_context`. The distractor probe runs on $N = 15$ `multi_turn_base` tasks where the opaque baseline already passes. For each task we remove the real target tool and inject one synthetic distractor in its slot. Appendix Table 6 reconciles which model saw which split: the “five Anthropic versions across eleven months” breadth lives on base only, the three-split regime sweep is two production models plus one local model, and the per-cell counts sum to the pooled 0/2,599 Anthropic figure.

Statistics. TFR is per parsed tool call. Per-cell rates get Clopper–Pearson upper bounds; non-zero cells get a BCa cluster bootstrap (10,000 resamples, clustered by task). Pooled C_0 vs. C_1 uses Fisher’s exact one-sided. Distractor-type effects use Friedman with Nemenyi post-hoc, and non-inferiority uses TOST at a 5-pp margin. We apply Holm–Bonferroni across the three primary tests.

Reproducibility. A `repro_manifest.json` pins the BFCL v4 commit, SDK versions, dated model aliases, decoding parameters, and random seeds (master seed 42). The supplementary archive at anonymous.4open.science/r/icml-tehr-scale-2026 holds the harness, 144 unit tests, the JSONL trace logs, and the scripts that regenerate every numerical claim (Appendix B).

5. Results

One finding, two regimes.

Table 1. Regime-test TFR by model and BFCL split under C_0 . Cells are events / parsed calls.

Model	base	miss_func	long_ctx
Sonnet 4.6	0/116	0/128	0/81
Haiku 4.5	0/109	0/120	0/107

Table 2. Anthropic 4.x distractor probe with target removed (C_0). Pooled C_0 over regime + probe is 0/2,599, $\leq 0.115\%$; the C_1 subset adds 0/678 at $\leq 0.441\%$.

Model	N	TFR
Opus 4.7	293	0
Sonnet 4.6	561	0
Sonnet 4.5	285	0
Sonnet 4	253	0
Haiku 4.5	525	0
Pooled probe	1,917	0

5.1. Regime tests: TFR is at or near zero across BFCL multi-turn splits

Per-call TFR for Sonnet 4.6 and Haiku 4.5 across the three BFCL multi-turn splits under C_0 is in Table 1. Every cell in the regime grid is **0/N** (C_0 calls only). Clopper–Pearson 95% per-cell upper bound is $\leq 3.63\%$; pooled over the 661 C_0 regime calls, $\leq 0.45\%$.

BFCL designed the `miss_func` split to elicit fabrication by removing a needed function. Both Anthropic models route around the missing tool by chaining smaller calls instead (one trace in Section 5.4). Qwen3-8B does not behave the same way. On the same split it logs $5/258 = 1.94\%$, slightly above the 1.46% on the aligned `multi_turn_base` split. That is what BFCL’s design predicts when the registry is deliberately incomplete.

5.2. Anthropic stability check across 5 model versions

The probe injects one synthetic distractor tool into a registry from which the real target tool has been removed: the agent expects a tool to exist, and the registry offers only a similarly-named distractor. Four distractor types (near-name, synonym, matched-random, unrelated) are tested on $N = 15$ `multi_turn_base` tasks where C_0 already passes. We ran five Anthropic 4.x versions over an 11-month window across the Opus, Sonnet, and Haiku families. On this probe every cell is **0/N** (Table 2; 95% pooled upper bound $\leq 0.16\%$). Combine the regime cells with the probe cells and the C_0 Anthropic audit pools to **0/2,599** (per-call aggregator, deduplicated across overlapping run-ids), or $\leq 0.115\%$.

On this benchmark, Anthropic models do not invent

Table 3. Qwen3 scaling curve under C_0 (target removed). Cells are events / parsed calls; bold marks the per-row peak.

Size	near	syn.	match.	unrel.	Pooled
0.6B	0/18	0/20	0/18	0/19	0/75
1.7B	2/54	0/54	0/49	0/53	2/210
4B	0/57	0/54	3/56	0/59	3/226
8B	1/219	3/134	3/129	2/133	9/615
14B	0/63	5/69	0/67	0/69	5/268
32B	0/55	0/57	0/55	0/57	0/224

tool names. Five versions, eleven months, the same null. Two readings fit the data and we cannot separate them: post-training hardened against fabrication, or BFCL’s redundant tool basket lets the agent route around a missing tool rather than guess at one. Honestly, both probably help. The gap with prior work is less mysterious once you remember that older models had no tool-use post-training and that prior benchmarks aggregated per task rather than per call.

5.3. Qwen3 scaling curve: TFR is non-monotonic across 0.6B–32B

The 14B→32B drop, 1.87% to 0%, is our central intra-family result (Table 3; plotted in Appendix Figure 2). Pooled per-call TFR rises from 0% (0/75, $\leq 3.92\%$) at 0.6B, peaks at 14B (5/268 = 1.87%), and collapses back to 0% (0/224, $\leq 1.33\%$) at 32B. That 32B upper bound is wide enough to overlap the 14B point estimate, so “collapse” here should be read as “not-detected at this N ,” not as emergent capability. The per-arm peak also shifts with scale (bold cells, Table 3): near_name at 1.7B, matched_random at 4–8B, synonym at 14B.

A same-family check strengthens the picture. Qwen2.5-7B-Instruct on the same probe gives 28/459 = 6.10%, higher than any Qwen3 size at any arm. This is not an apples-to-apples generation comparison (different post-training, different tokenizer), but it says the failure mode is not a Qwen3-specific artefact of one release. The lineage carries it.

Commercial models track the Anthropic null rather than the Qwen3 band. On the same probe, the OpenAI gpt-4.1 and gpt-4o tiers also log 0% TFR: gpt-4.1 0/404, gpt-4.1-mini 0/408, gpt-4.1-nano 0/409, gpt-4o 0/419, gpt-4o-mini 0/430 (per-model Clopper–Pearson 95% upper bounds ≤ 0.69 – 0.74%), pooling to 0/2,070 at $\leq 0.14\%$. The cross-vendor pattern is an open-vs-closed split: zero-event commercial frontiers against the Qwen-family 0.95–1.87% band.

We treat the shape itself as the result. Most read-

Table 4. RVR per-tier validation on Qwen3 sizes with non-zero C_0 events. Cells are events / parsed calls. These are the paired C_0/C_1 matched-run cells; their C_0 denominators (e.g. 8B 4/269) are smaller than the full distractor-probe pool in Table 3 (8B 9/615).

Tier	C_0	C_1	Prevented
Qwen3-1.7B	2/210	0/200	2/2
Qwen3-4B	3/226	0/228	3/3
Qwen3-8B	4/269	0/258	4/4
Qwen3-14B	5/268	0/259	5/5
Pooled	14/973	0/945	14/14

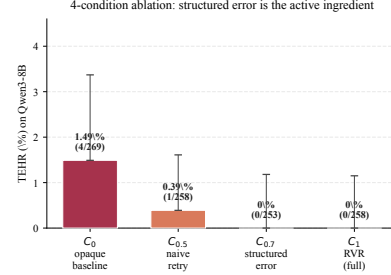


Figure 1. Qwen3-8B 4-condition ablation. The structured-error envelope ($C_{0.7}$) drives the prevention; adding the registry list (C_1) adds no further reduction.

ers would guess that 14B fabricates less than 1.7B. It does not. The traces suggest why. At 1.7B the model rarely emits well-formed tool calls, leaving little surface to fabricate on. At 14B it formats calls fluently but does not check the registry. At 32B it does both. Six points do not support a fitted scaling law, but the shape is consistent: at each scale the dominant failure is whichever distractor that scale is best primed to mimic.

5.4. RVR per-tier validation: 14 → 0 events across the Qwen3 sweep

We applied the registry-visible re-prompt C_1 on the four Qwen3 sizes with TFR > 0 in C_0 . Across ~ 944 matched calls, RVR prevents all **14** of **14** hallucination events (Table 4).

The Anthropic C_1 probe (Sonnet 4.6 + Haiku 4.5) preserves 0/539 hallucinations. The strict-pass subset drops: Sonnet 60/60 → 53/60 (−11.7 pp), Haiku 60/60 → 56/60 (−6.7 pp). A Newcombe-hybrid TOST at the 5 pp margin gives $p_{\text{TOST}} = 0.91$ (Sonnet) and 0.63 (Haiku). At $N = 60$ the test is underpowered ($1 - \beta < 0.30$), so we can call neither non-inferiority nor inferiority. RVR stays zero-event on the Anthropic frontier; the strict-pass cost is undetermined at this N (Section 7).

The four-condition ladder on Qwen3-8B walks down

a clean gradient (Figure 1). Opaque baseline 1.46% (9/615), naive retry 0.39% (1/258), structured envelope $C_{0.7}$ 0/448 ($\leq 0.67\%$), full RVR C_1 0/258. The structured envelope drives the prevention; stacking the registry list on top buys no further reduction (Newcombe-hybrid 95% CI on $C_1 - C_{0.7}$: $[-1.5, +1.4]$ pp), as we analyze in Section 6. The practical upshot: a deployer who wants prevention without echoing internal tool names can ship $C_{0.7}$ and lose nothing.

Fisher’s exact on the pooled $\text{TFR} > 0$ subset. The C_0/C_1 denominators differ per tier, so the trials are not paired at the call level; we treat the four tiers as independent strata. We report Fisher’s exact one-sided test on the pooled 14/973 vs. 0/945 contingency: $p = 7.1 \times 10^{-5}$. Per-tier Fisher’s exact one-sided p values are $\{0.24, 0.12, 0.058, 0.032\}$ for 1.7B/4B/8B/14B respectively. Each tier is individually underpowered, but the sign is consistent. The pooled contrast survives Holm–Bonferroni at $p_{\text{Holm}} = 2.14 \times 10^{-4}$.

Trace inspection. On `multi_turn_miss_func_24` the Anthropic agent chains five in-registry calls (`pwd`→`ls`→`cd`→`mkdir`→`mv`) where a one-shot `mv_recursive()` would have been fabricated. Across 30 near-name probe traces, agents reach for other in-registry tools rather than the removed target’s or the distractor’s name. RVR adds zero provider round-trips on hits, one on misses; the rendered registry is ~ 100 –200 tokens at ≤ 25 tools (under 5% of context).

6. What recovers: content or format

Does recovery use the content of the echoed registry, or only the format of a structured rejection? The condition ladder answers this directly, without the distractor matrix we are underpowered to read.

The ablation already points away from content. On Qwen3-8B the structured error $C_{0.7}$, a `tool_not_found` envelope with no registry attached, already drives fabrications to zero, matching full RVR (Section 5.4). If the model needed to read the registry to recover, removing it should cost something; it costs nothing.

The decoy arm closes the question. $C_{0.7}$ removes the list; it does not test what happens when the list is present but wrong. A registry-as-cue account predicts a decoy list (correctly formatted, count-matched, but populated with non-existent tools) should fail to recover, with no valid name to copy. A format-only account predicts it recovers anyway, since the constraint

“your last call was rejected; choose from this set” need not be truthful to suppress the out-of-registry call. The $C_{0.8}$ condition runs exactly this contrast: the C_1 envelope prose held verbatim, only the listed names swapped for guaranteed-absent decoys.

The decoy list recovers at 0/410 (Clopper–Pearson 95% upper bound $\leq 0.73\%$) across all four probe arms (near_name 0/97, synonym 0/106, matched_random 0/107, unrelated 0/100), indistinguishable from the real list (C_1 , 0/258 on 8B) and from the no-list envelope ($C_{0.7}$, 0/448). A wrong list recovers as well as the right one: registry content is decorative for recovery. The model is responding to the message-level rejection itself, not being cued by valid names; the truthfulness of the offered set is not load-bearing. This is positive evidence against the registry-as-cue account, which would need a valid name to point at and the decoy offers none.

What this does not claim. Format-sufficiency is a claim about recovery, not task success. A decoy list suppresses the fabricated call but offers no valid target, so where the real tool was removed the model must still decline or redirect; the value of the true registry shows up in task-completion rate, not in TFR, which is the quantity the rejection signal governs. Nor does format-sufficiency hold at every scale: it is established only on the tiers where C_0 fabrications exist to be suppressed (1.7–14B); the 0.6B and 32B tiers carry no signal to recover.

The scale curve, descriptively. The dominant distractor type drifts with scale (Section 5.3), but a Friedman exact permutation test is underpowered ($\chi^2_{(3)} = 3.0$, $p = 0.46$; $n = 4$ sizes), so we read the drift as phenomenology, not inference. The content-vs-format question rests not on this matrix but on the powered $C_{0.7}$ and $C_{0.8}$ ablations.

7. Discussion and Limitations

We cannot disentangle the Anthropic–Qwen3 gap: parameter count, 4-bit MLX quantization, training recipe, and provider guardrails co-vary, and the non-monotone Qwen3 shape rests on one family at one quantization. The Anthropic zero-event regime is indistinguishable from BFCL v4 contamination, since v4 predates every model tested and we ran no paraphrase or membership-inference control. RVR’s 14→0 prevention is invariant to quantization, as C_0 and C_1 share weights. On Anthropic, RVR is a call-path no-op yet strict-pass drops (Section 5.4), and our Newcombe-hybrid TOST is underpowered at $N =$

60. We recommend $C_{0.7}$ /RVR only on tiers with audited $TFR > 0$, paired with executor-side allow-listing. Scope: the probe tests registry-shape, not runtime-shape (Czapla’s `read_secret()` collision needs ambient-reachability BFCL lacks); TFR ignores latency, abstention, and pass-rate; and we claim no scaling law or cross-family transfer.

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LLM/Agent Usage Disclosure

Experimental subjects. The paper evaluates eleven LLMs as subjects of measurement: five Anthropic 4.x versions (Opus 4.7, Sonnet 4/4.5/4.6, Haiku 4.5) via Anthropic’s tool-use API, and six Qwen3-Instruct sizes (0.6B–32B) as 4-bit MLX builds on a 32 GB Apple Silicon machine. All are treated as black-box targets; version snapshots, dated aliases, and decoding configurations are pinned in the supplementary manifest.

Drafting, code, and statistics. The paper text and the harness (adapters, intervention conditions, statistical scripts, trace logger, cost meter) were drafted with LLM-based assistance under author supervision. All claims, numerical results, methodological decisions, and statistical tests (Fisher’s exact, Newcombe-hybrid TOST, Holm–Bonferroni, BCa cluster-bootstrap, Friedman+Nemenyi) are reproducible from the released harness and its 144 unit tests; no p -value appears without a code path that regenerates it. Contributions, experimental design, scope claims, and limitations were authored under direct human supervision.

A. Supplementary Figures and Tables

B. Pre-Registration and Reproducibility

The pre-registered protocol is locked in git commit c391017, dated before any data was collected. Three deviations from that protocol are worth naming. Probe N was reduced from 100 to 15 per cell to fund breadth across five Anthropic versions and six Qwen3 sizes. The OpenAI gpt-4.1 and gpt-4o tiers were added to the vendor scope after the pre-registration (five commercial models, 0/2,070 pooled C0 calls, all 0% TFR);

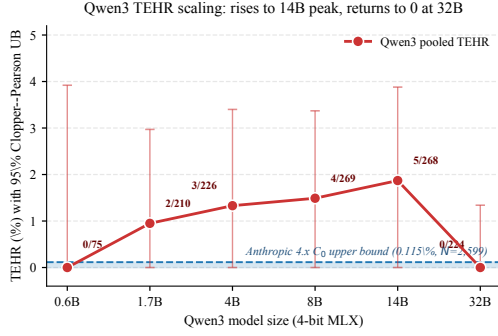


Figure 2. Qwen3 family TFR scaling curve, opaque baseline C_0 , four-distractor probe with target removed. Error bars are Clopper–Pearson 95% CIs; the dashed blue Anthropic 4.x upper bound pools all 5 tested versions. The intra-family non-monotone shape and the gap between the open-source ~ 2 –14B band and the Anthropic bound is the paper’s headline.

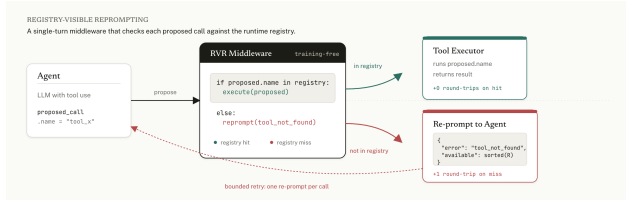


Figure 3. RVR middleware. An $O(1)$ membership check runs in-process on each proposed call. Hits execute. Misses trigger one re-prompt with a structured `tool_not_found` envelope (and the sorted registry under C_1). The no-violation path adds zero provider round-trips; the violation path adds one, bounded to a single retry.

the gpt-5 family is still in collection. $C_{0.7}$ was run at scale on Qwen3-8B only, not on Anthropic. Holm–Bonferroni correction across the three primary tests (Fisher’s exact, Friedman, TOST): only Fisher’s clears $\alpha=0.05$ at $p_{\text{Holm}} = 2.14 \times 10^{-4}$.

The supplementary repository at anonymous.4open.science/r/icml-tehr-scale-2026 contains the full harness (144 unit tests), the JSONL trace logs that produce every number in the paper, the aggregator and statistical scripts, the reproducibility manifest with dated model aliases, the full deviation log, the JSONL trace schema, the headline-number derivation rule, and a registry-size pilot. Running `python scripts/aggregate_all.py` on the released traces reproduces every numerical claim. Distractor-injection RNG uses a SHA-256-derived stable hash, so reproduction does not depend on PYTHONHASHSEED; MLX 4-bit local inference is not bitwise-deterministic across runs but per-task tool-call decisions are stable to within ± 1 token.

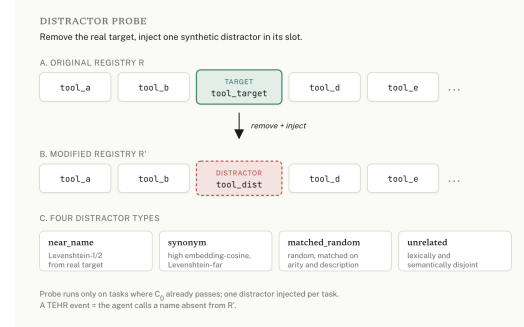


Figure 4. The distractor probe. Starting from a registry where the opaque baseline already passes, the real target tool is removed and one synthetic distractor is injected in its slot. The four arms vary the similarity profile of the distractor.

Table 5. Prior tool-hallucination measurements versus TFR. Granularity is the denominator each metric divides by; existence-clean? asks whether the metric isolates registry-membership violations (a called name is absent from the offered set) from relevance, argument, and timing errors. Not apples-to-apples: the headline numbers come from different denominators, different model cohorts, and different elicitation setups (single-turn instructions vs. multi-turn agentic traces; target-removed probes vs. natural distributions). They are not directly comparable and we do not claim a ranking; the table positions what each metric counts, not which model is best. TFR is the only per-call rate and the only one that scores existence on multi-turn traces with a content-controlled ablation.

Work	Granularity	Setting
API-Bank (Li et al., 2023)	per-task	single-turn API calls
MetaTool (Huang et al., 2024)	per-query	single-turn, target removed
ToolBeHonest (Zhang et al., 2024)	per-task	multi-level diagnostic
Reasoning-Trap (Yin et al., 2026)	per-task	no-tool / distractor-only
Reliability align. (Xu et al., 2024)	composite	training-time, expanded
Ours (TFR)	per-call	multi-turn agentic traces

Table 6. Model $\times$ BFCL-split coverage. Cells are no-intervention (C0) calls; a glyph marks the experiment design that produced them. $\bullet$: the five-distractor probe grid (target removed, per-call existence scored). $\circ$: the regime cell from the main grid. The “five Anthropic versions across eleven months” claim holds on base only; the three-split regime sweep is two production models (Haiku 4.5, Sonnet 4.6) plus one local model (Qwen3-8B, base). Anthropic hallucinated = 0 in every cell; the per-call tehr signal is the Qwen3 size sweep on base.

Model (version)	BFCL split (C0 calls)		
	base	long-ctx	miss-func
Anthropic 4.x			
Sonnet 4.0 (05/2025)	$\bullet$ 253	–	–
Sonnet 4.5 (09/2025)	$\bullet$ 285	–	–
Sonnet 4.6	$\bullet$ 677	$\circ$ 81	$\circ$ 128
Opus 4.7	$\bullet$ 293	–	–
Haiku 4.5	$\bullet$ 655	$\circ$ 107	$\circ$ 120
Qwen3 size sweep (4-bit)			
0.6B	$\bullet$ 75	–	–
1.7B	$\bullet$ 210	–	–
4B	$\bullet$ 226	–	–
8B	$\bullet$ 615	$\circ$ 108 (base regime)	–
14B	$\bullet$ 268	–	–
32B	$\bullet$ 224	–	–